

What is artificial intelligence?

Understand the idea. Look inside the system.

Learn when it belongs in your work.

🕒 20-25 min core reading · Beginner friendly · 16 short sections

AI is bigger than a chat window. An email filter, an image recognizer, and a personal learning assistant solve different problems. This chapter explains what connects them, how they work at a high level, and when to use them.

Your destination

Explain AI in your own words, distinguish rules from learning, trace an AI application's decisions, and choose a sensible approach to a practical problem.

How to study: Allow about 20-25 minutes for the core reading and diagrams. Try the experiment and three core exercises afterward. Expand optional notes, the Python lab, and extra practice when useful. No AI background or advanced mathematics is needed.

1. Begin with a problem, not a technology

Imagine an inbox receiving 150 emails a day. You want to catch spam while keeping legitimate messages easy to find. A phrase rule or a learned spam classifier might help. Choosing AI is a method decision; first define the problem.

Question	Email-filter example
Input: What information is available?	Message text and available sender information
Task: What should the system do?	Identify likely spam
Output: What should it return?	A spam score and a flag-or-accept decision
Success: How will we judge it?	Catch spam while limiting false alarms on legitimate messages

“Use AI” is vague. “Catch spam without hiding important messages” gives us two measurable goals. If a simple rule solves the problem reliably, a model may be unnecessary.

2. What does “artificial intelligence” mean?

Artificial intelligence is the field of building computational systems that perform tasks involving perception, language, reasoning, planning, learning, or decision-making. This is a working definition; the field’s boundaries are debated. See [Poole and Mackworth’s introduction](#) for an agent-oriented perspective.

AI includes search, logical rules, and learning methods. **Machine learning (ML)** is an approach within AI that fits models using data. The two terms are not interchangeable. [Google’s ML introduction](#) explains this learning approach.

Capability is a demonstrated ability on a task. **Reliability** is how consistently the system meets requirements across relevant situations. Recognizing some photographs does not prove an ability to plan trips-or even reliable recognition of every kind of photograph.

▼ Optional perspective: does AI think like a person?

An aircraft flies without flapping its wings like a bird. Similarly, performing an intelligent-looking task does not establish human-like thought, consciousness, or feelings. Judge the demonstrated behavior and its limitations.

Chapter 2 will map the major AI terms and their relationships.

3. Look inside an AI application

A spam-classification model is one component. The complete application must receive messages, prepare their text, obtain scores, and decide which messages to flag.

➤ An AI application has several responsibilities

FROM INFORMATION TO A USEFUL RESULT



Figure 1. A model output becomes useful through preparation, application policies, and review. Feedback can inform later improvements; it does not automatically retrain the model.

Component	Role in the application
Data and preparation	Receive information and check or organize it before use
Algorithm	A specified computational procedure, such as a method for fitting a model
Model	A representation used to produce outputs; an ML model contains state fitted using data
Application logic	Apply policies, such as sending uncertain cases for review
Interface	Let people supply inputs and inspect results
Evaluation	Compare results with defined requirements

The training algorithm is the **procedure**; the fitted model is the **resulting state used for predictions**. A model might output a spam-like score of 0.82. Moving a message to a spam folder is a separate action, controlled by application policy. A score alone does not grant permission to act.

4. The different kinds of work AI can do

These capabilities can be combined in one application:

Capability	Meaning and example
Perceive	Extract information from sensory input, such as recognizing a handwritten digit
Classify	Assign a category, such as an email label, “spam”
Estimate	Produce a number, such as a predicted house price
Rank	Order candidates, such as search results ordered by relevance
Generate	Produce content, such as a summary of a study note
Reason or solve	Use relationships or constraints, such as checking whether two lessons fit a study session
Plan	Choose steps toward a goal, such as finding a robot’s route through a maze
Learn	Adjust a model from data or experience

Three distinctions prevent confusion:

Prediction is broader than forecasting. Identifying the language of an existing sentence predicts a label; forecasting estimates something in the future.

Receiving input is not perception. Loading an image reads data; recognizing its objects interprets that data.

Planning is not execution. Proposing a study schedule and adding events to a calendar are different operations.

5. Explicit rules versus learned patterns

A developer can write a spam rule directly:

```
def should_flag(message):  
    return "claim your prize" in message.lower()
```

This catches "CLAIM YOUR PRIZE now." It misses different wording such as "A reward awaits; send your account details." It also flags a legitimate warning that quotes "claim your prize." The rule follows a phrase, not the message's intent.

In **supervised learning**, a training procedure uses examples paired with target answers, called **labels**. It adjusts model **parameters**-internal numerical values-to improve outputs against an objective. **Features** are input properties represented for the model, such as word counts or a numerical representation of the message.

➤ Two ways to specify behavior

EXPLICIT RULES



LEARNED PATTERNS



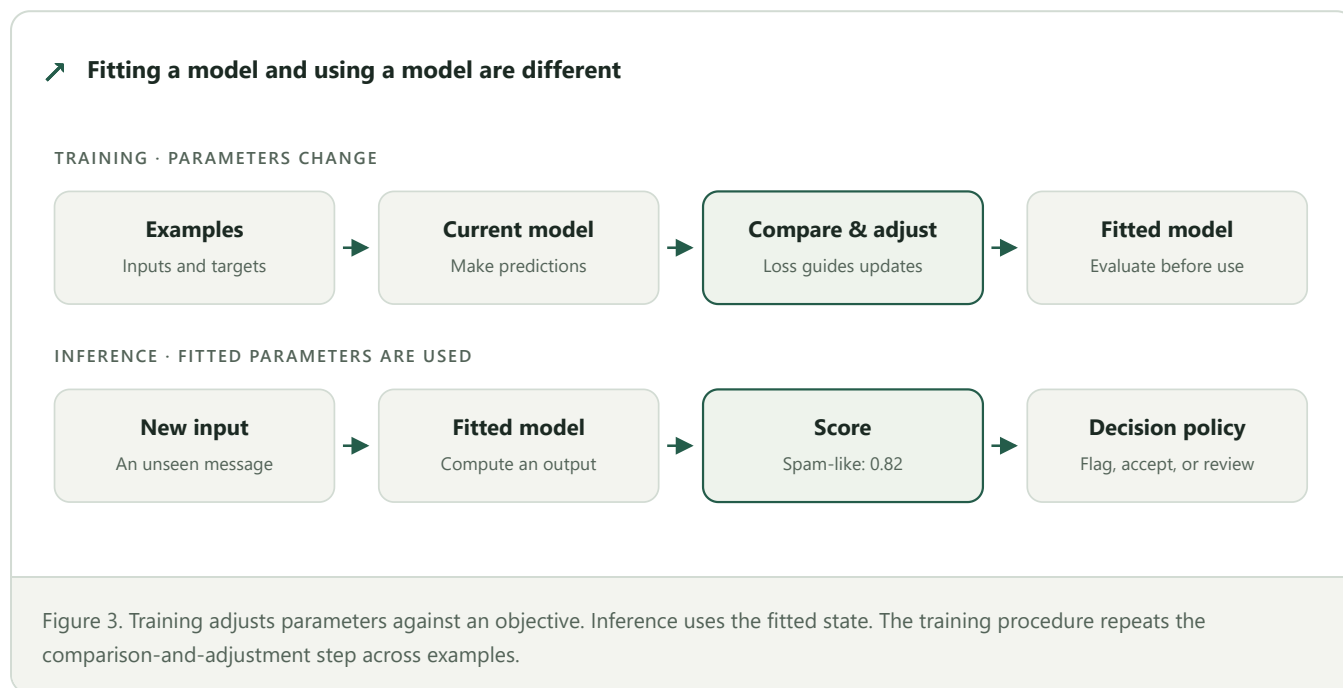
Figure 2. Explicit rules encode chosen conditions directly. Learning fits a model from examples. Both approaches still require evaluation.

Rules provide directly specified behavior; learned models can capture patterns that are difficult to list manually. Both can fail and need testing. People still choose the data, model family, training procedure, and evaluation method.

Do not equate rules with “never AI.” Classical AI includes rule-based expert systems and search. A simple phrase filter is usually called automation; reasoning over a knowledge base may be called symbolic AI. Ask which parts are specified, which are learned, and how each is checked.

6. Training and inference: two different moments

Training fits model parameters using data. **Inference** runs the fitted model on an input. Here, “inference” means model execution, not necessarily logical deduction.



During training, a **loss** measures how poorly outputs satisfy the training objective. The learning procedure uses it to guide parameter updates. During inference, many deployed models keep their parameters fixed while producing outputs for new inputs.

A student who memorizes practice answers may fail unfamiliar questions. Likewise, **generalization** means useful performance on relevant new cases. Evaluate on examples that were not used to fit the model.

Overfitting: fitting training data so closely that performance on new cases suffers.

Distribution shift: incoming data differs from development data—for example, messages arrive in a previously untested language.

New context and feedback are not automatically training. Supplying a document changes the information available for a response. Correcting a category may merely record feedback. Whether either updates the model depends on the application.

7. Worked example: a score becomes a decision

A **threshold** is a cutoff that turns a score into a decision. Our rule is: **flag a message when its score is greater than or equal to the threshold.**

The eight messages below have synthetic labels and fixed illustrative scores. No model was trained for this demonstration. Higher scores mean “more spam-like,” but they are **not established probabilities**: 0.80 does not automatically mean an 80% chance of spam.

Interactive experiment 01

Synthetic data · no training

Same scores. Different decisions.

Move the cutoff. Notice which mistakes appear and disappear.

2/4

Spam caught

1

False alarms

2

Spam missed

62.5%

Accuracy

A	Claim a cash prize Actual: spam		0.96	Spam caught
B	Pay a release fee Actual: spam		0.88	Spam caught
C	Subscribed newsletter Actual: legitimate		0.73	False alarm
D	Unknown reward link Actual: spam		0.67	Spam missed
E	Registered workshop Actual: legitimate		0.54	Correctly accepted
F	Private opportunity Actual: spam		0.45	Spam missed
G	Meeting agenda Actual: legitimate		0.21	Correctly accepted
H	Requested report Actual: legitimate		0.08	Correctly accepted

The line marks your threshold. The bars are fixed scores, not guaranteed probabilities. Flagging is a policy choice; moving this slider does not train a model.

Print snapshot: at 0.70, A, B, C are flagged; 2 spam caught, 1 false alarm, 2 spam missed. At 0.40: 4 caught, 2 false alarms, 0 missed.

At the default threshold of **0.70**, A, B, and C are flagged:

Result	Messages	Meaning
Correct spam detection	A, B	Spam is caught
False positive	C	A legitimate message is flagged
False negative	D, F	Spam is missed
Correct acceptance	E, G, H	Legitimate messages are left unflagged

There are five correct decisions out of eight:

$$\text{Accuracy} = \frac{\text{correct decisions}}{\text{all decisions}} = \frac{5}{8} = 62.5\%$$

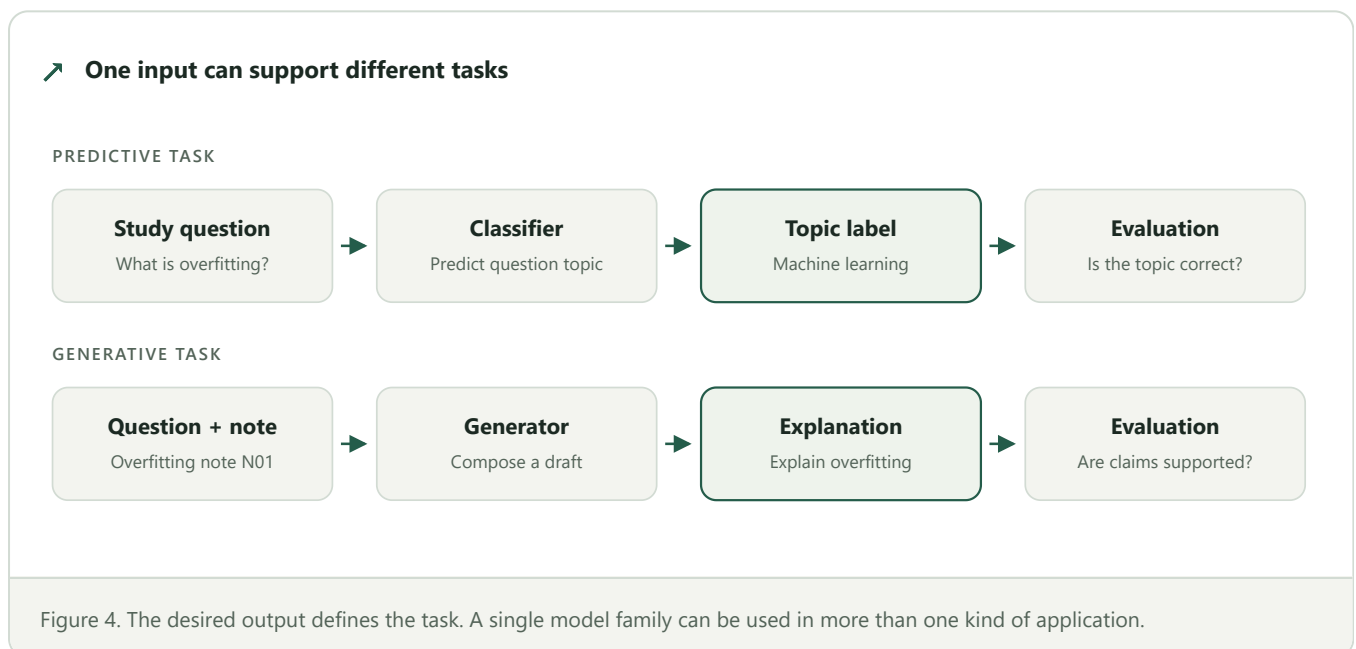
Dividing 5 by 8 gives 0.625; multiplying by 100 expresses it as a percentage. Accuracy counts correct decisions but does not distinguish the costs of different mistakes. [More on classification measures.](#)

Try 0.40: all four spam messages are caught, but two legitimate messages are flagged. Six decisions are correct: 75% accuracy on this tiny dataset. At 0.90, only A is flagged, so more spam is missed.

Only the policy changed; the scores did not. Select a real application's threshold using representative validation examples and the costs of false alarms versus missed cases-not this teaching dataset.

8. Predictive tasks and generative tasks

Predictive tasks produce estimates, categories, scores, or rankings. **Generative tasks** produce content, such as text, images, audio, or code.



Consider a question submitted to a **personal learning assistant**: "What is overfitting?" It has this short fictional course note:

No1 • Overfitting: A model can fit training examples too closely and perform poorly on new cases. This lesson has an estimated study time of 20 minutes. The notes do not list an exam date.

A predictive classifier might assign the question the topic **machine learning**. A generator might compose an explanation:

Supported draft: “Overfitting means fitting training examples so closely that performance on new cases suffers [N01]. The note does not state your exam date.”

Unsupported draft: “Your exam is Friday, and studying for 20 minutes guarantees mastery.” Neither the date nor guaranteed mastery follows from N01.

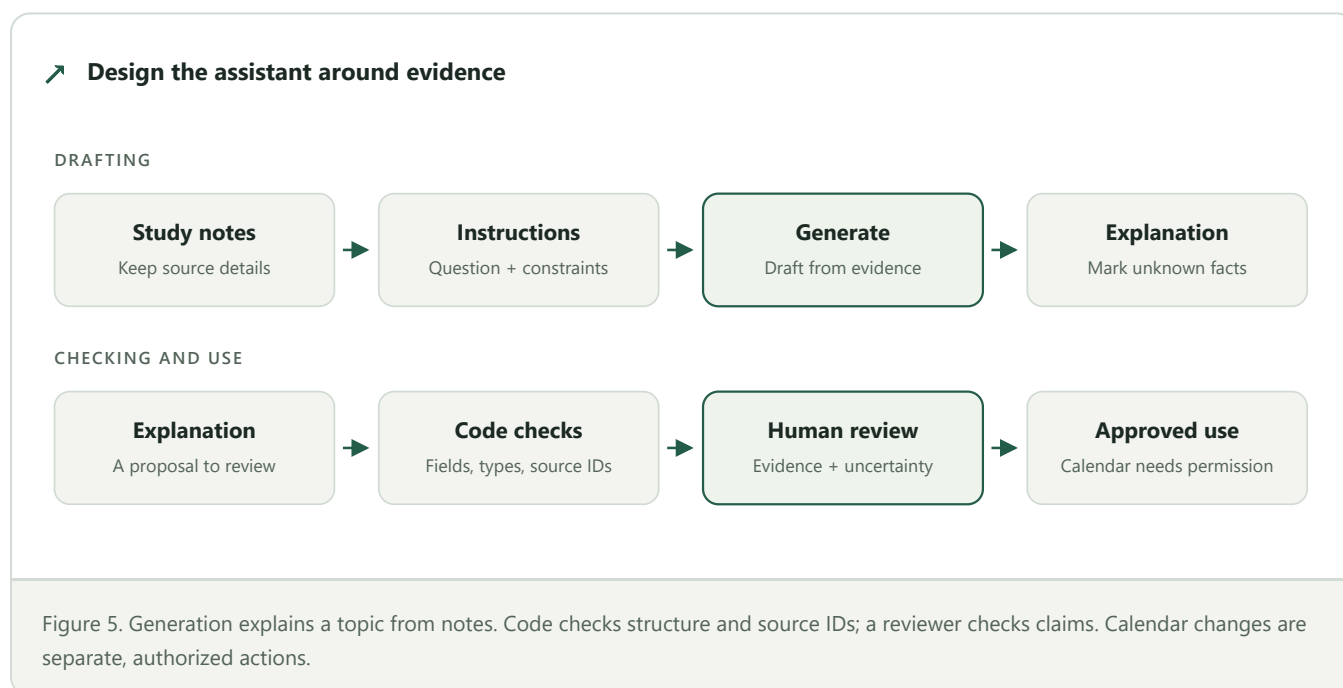
Grounding ties an answer to supporting information. A **hallucination**, here, is incorrect or unsupported generated content presented as an answer. Check support, important omissions, and preserved meaning-not just fluent writing.

▼ Optional connection: can generation also involve prediction?

Yes. A language model often generates text by repeatedly predicting a next **token**, a unit of text represented by the model. Its internal operation involves prediction; the application’s output is generated content. The same model can also output one category label. Model mechanism and application task describe different levels.

9. Real-world walkthrough: a personal learning assistant

Our goal is a clear explanation from the learner’s notes, with sources and any missing information identified. Here is how the components work together:



Specify the result. Answer the learner’s question clearly; separate what the notes say from what remains unknown.

Supply evidence. Provide relevant notes, the question, and source references such as N01. **Retrieval-augmented generation (RAG)** supplies retrieved information to support an answer; we will build it later.

Generate a draft. Request the required sections and explicit treatment of missing facts.

Validate. Code checks required fields, source identifiers, and calculations. These checks do not prove factual accuracy.

Review claims. A person checks the draft against evidence before approved use.

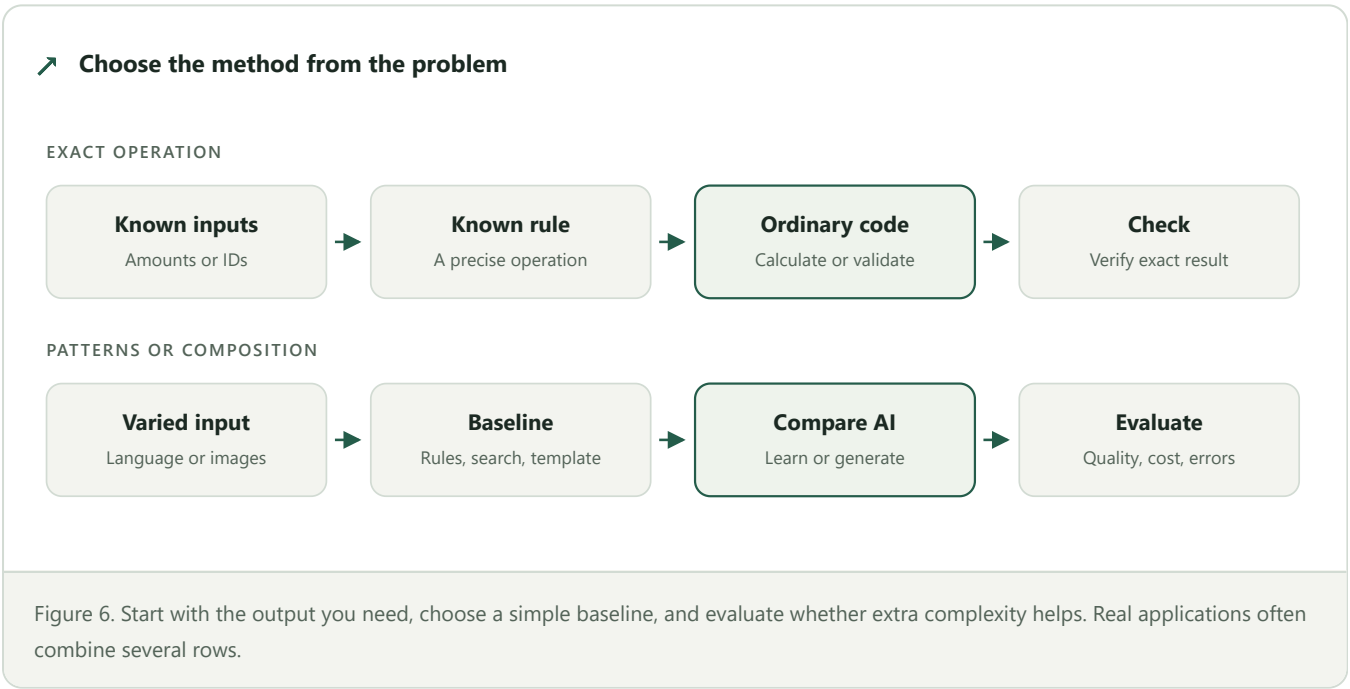
Improve deliberately. Use reviewed errors to revise prompts, data handling, or models; then evaluate on separate examples.

A calculation can be correct while its interpretation is wrong. A 30-minute study session minus N01’s estimated 20 minutes leaves 10 minutes. That establishes a time budget; it does not establish that the learner will master the topic in that time.

Students who revise more might also score higher on exams. **Correlation** is an association; **causation** is a claim that changing one factor affects another. That association alone does not prove that adding a particular revision session causes a higher score: prior knowledge and study methods may also matter.

10. When should you use AI?

A **baseline** is a simple reference solution used for comparison. Start there and add complexity when it measurably helps.



Need	Sensible first approach
Add events to a study calendar	Authorized application logic and an explicit approval policy

Before choosing, ask: **Is the task clear? Is the evidence available? What does an error cost? Can we measure usefulness? Would a simpler method suffice?** Include verification effort: faster drafting is not a gain if checking the draft takes longer than doing the original task.

11. Limitations and misconceptions

Misconception	Better understanding
Confident wording proves correctness	Tone is not evidence; verify claims
More data always helps	Irrelevant data, wrong labels, duplicates, or missing groups can reinforce errors
A learned model must vary its answer	Many are deterministic: the same input and model state can give the same output; sampling can introduce variation
One successful example proves readiness	Test ordinary, difficult, incomplete, and unanswerable cases
A good average score is enough	Inspect the error types and whether the test resembles real use

Use public or synthetic data for initial practice. For workplace applications, use appropriate data permissions and an approved environment. Models still need the surrounding validation, access controls, and review introduced above.

12. Optional Python lab: inspect a simple baseline

The phrase rule from Section 5 can be checked against four examples. It gets two right, misses differently worded spam, and flags a legitimate warning. **The labels check the rule; they do not train it.**

▼ Optional lab: code, output, and a small experiment

Run this with Python 3; no additional packages or account are needed.

```
def should_flag(message):
    return "claim your prize" in message.lower()

examples = [
    ("CLAIM YOUR PRIZE now", True),
    ("A reward awaits; send your account details", True),
    ("Training: avoid messages saying 'claim your prize'", False),
    ("Tomorrow's project meeting agenda", False),
]

correct = 0
for message, expected in examples:
    predicted = should_flag(message)
    correct += int(predicted == expected)
    print(f"predicted={predicted}, expected={expected}")

print(f"Correct decisions: {correct}/{len(examples)}")
```

Expected output:

```
predicted=True, expected=True
predicted=False, expected=True
predicted=True, expected=False
predicted=False, expected=False
Correct decisions: 2/4
```

`int(True)` is 1 and `int(False)` is 0, so the counter increases for correct results. `len(examples)` counts the examples.

Add another trigger phrase, predict what changes, then test new messages-including a legitimate message quoting that phrase. A broader rule can catch more spam and create more false alarms. These four cases demonstrate behavior; they do not estimate real-world performance.

13. Practice: explain, decide, and investigate

Do these three core exercises first. Explain your reasoning before revealing an answer.

Exercise 1: explain it without buzzwords

Explain AI in three sentences. Include a task, a method, and a limitation.

▼ Show a sample answer

AI builds systems for tasks such as recognizing information, using language, and choosing actions. A model fitted on labeled emails can score new messages for spam. Its errors must be evaluated because unfamiliar cases or misleading learned patterns can flag legitimate mail or miss spam.

Exercise 2: calculate the mistakes

Set the eight-message experiment to **0.60**. Which messages are flagged? Count false positives, false negatives, and correct decisions.

▼ Show the calculation

A, B, C, and D are flagged. A, B, D are correctly detected spam; C is one false positive. F is one false negative. E, G, H are correctly accepted. Six of eight decisions are correct: 75% accuracy.

Exercise 3: check the evidence

A draft claims: “Your exam is Friday, and studying N01 for 20 minutes guarantees mastery.” What can the course note in Section 8 actually support?

▼ Show the evidence check

N01 supports the definition of overfitting and a 20-minute study-time estimate. It does not establish an exam date or guarantee mastery. Explain the concept using the note, label the time as an estimate, and keep the exam date unknown.

▼ Optional practice: five more exercises with solutions

Exercise 4: choose the first approach

Choose an approach for adding amounts, detecting an absent ID, grouping study questions, and explaining a topic.

▼ Show the reasoning

Use arithmetic for amounts and a validation rule for an absent ID. For study questions, clarify whether groups are predefined topics or exploratory groupings, then compare a simple baseline. Use a template or generative explanation for the topic, checking claims against the supplied notes.

Exercise 5: distinguish training from inference

A model is fitted on 2,000 labeled messages, categorizes 50 new ones, and receives one user correction. What happened at each stage?

▼ Show the answer

Fitting is training; categorizing new messages is inference; the correction is feedback. It updates the model only if the application includes an update or retraining process.

Exercise 6: question an impressive score

Only two of 100 messages are spam. A system calls every message legitimate. How accurate is it, and what does that hide?

▼ Show the calculation

It is 98% accurate but misses both spam messages. Evaluate detection and false alarms separately; accuracy can hide failure on an infrequent class.

Exercise 7: sketch your own system

Choose a task. Identify its input, output, baseline, AI component, two failure cases, and checking method.

▼ Show a worked example

For a learning assistant, use a question and study notes as input and an explanation with sources as output. Compare generated explanations with search plus quoted passages. Watch for irrelevant notes and invented exam dates. Check claims against the sources and keep calendar changes subject to permission.

Exercise 8: explain the experiment

Does lowering the spam threshold make the model learn?

▼ Show the answer

No. The scores remain fixed. The cutoff changes the decision policy without refitting model parameters.

14. Check your understanding

Use this short quiz as a self-check. Read the explanations for any uncertain answers.

01 Which statement describes AI most accurately?

- A. Every program that automates a task is AI.
- B. AI includes methods based on search, rules, and learning.
- C. AI and chatbots mean the same thing.

Answer: B. AI is a broad field. Chatbots are one application, and simple automation alone does not establish an AI method. Rules and search can be part of classical AI.

02 A model assigns a score of 0.82. What can we conclude?

- A. The event has exactly an 82% chance of occurring.
- B. The application must automatically act.
- C. We need the score definition and decision policy to interpret it.

Answer: C. Scores need a defined meaning. Probability calibration is a separate property, and an action is chosen by an application policy.

03 Which operation is training?

- A. Adjusting model parameters using examples and an objective.
- B. Supplying a new document in the input.
- C. Moving a decision threshold from 0.70 to 0.40.

Answer: A. Training fits parameters. New context changes the information supplied, while changing a cutoff changes the policy around an output.

04 A study note defines overfitting but gives no exam date. What should the assistant say?

- A. The exam must be this Friday.
- B. Explain overfitting using the note; keep the exam date unknown.
- C. Reading the note guarantees mastery.

Answer: B. The note supports the definition. It does not establish an exam date or guarantee learning outcomes.

05 What is the best initial approach to adding known amounts?

- A. A large language model with no checking.
- B. A classifier trained on invoices.
- C. A deterministic calculation in code or a spreadsheet.

Answer: C. The arithmetic operation is precisely specified. Exact computation provides a clear, checkable baseline without needing a learned model.

15. Revision notes and glossary

AI is broader than ML: it also includes search, rules, and planning.

Define the task first: input, task, output, and success criteria.

Training fits; inference uses. New context, feedback, and threshold changes are separate mechanisms.

A model is one component. Policies turn its outputs into actions.

Evidence matters: check generated claims and test on relevant new cases.

Compare with a baseline: inspect different errors and their practical costs.

▼ Reference: mini glossary

Mini glossary

Term	Meaning
AI	Field covering capabilities such as perception, reasoning, language, and learning
Machine learning	Fitting models using data
Algorithm	Specified computational procedure
Model	Representation used to produce outputs; ML models contain fitted state
Parameter	Adjustable internal model value
Feature	Input property represented for a model
Label	Supplied target answer or category
Dataset	Collection of examples
Training / inference	Fitting a model / running it on an input
Loss	Measure of mismatch used in a training objective
Generalization	Useful performance on relevant new cases
Threshold	Score cutoff used for decisions
False positive / negative	Incorrectly flagged case / missed positive case
Baseline	Reference approach for comparison
Grounding	Tying an answer to supporting information
Hallucination	Incorrect or unsupported generated content presented as an answer
Token	Unit of text represented by a language model
Validation	Checking requirements; a validation dataset supports development decisions

Before moving on: Explain why the phrase rule is not learning, why lowering a threshold changes errors, and why the learning assistant cannot invent an exam date. Then draw an everyday application and label its input, AI component, ordinary code, review step, and evaluation method.

16. Sources and where we go next

Examples, data, diagrams, and exercises were created for this handbook. Optional further reading:

Poole and Mackworth: What is AI?

Google: What is Machine Learning?

Google: Classification measures

scikit-learn: Metrics and scoring

Next: Chapter 2 connects AI, ML, deep learning, NLP, generative AI, LLMs, RAG, agents, and AGI.